

Optimization

◆ Definition

Optimization is the process of **improving the performance, efficiency, or quality** of a system by making the best use of available resources while satisfying given constraints.

◆ Objectives of Optimization

- Maximize performance (speed, throughput)
 - Minimize cost, delay, or energy consumption
 - Efficient utilization of resources
 - Improve Quality of Service (QoS)
-

◆ Types of Optimization

1. Network Optimization

- Improves performance of communication networks
 - Example:
 - Reducing congestion
 - Improving signal strength
 - Load balancing
-

2. Code Optimization

- Improves program efficiency
- Performed by compiler or programmer

Types:

- **Machine-independent optimization** (loop optimization, dead code removal)
- **Machine-dependent optimization** (register allocation)

3. Database Optimization

- Improves query performance
 - Example:
 - Indexing
 - Query rewriting
 - Normalization
-

4. Resource Optimization

- Efficient allocation of:
 - CPU
 - Memory
 - Bandwidth
-

◆ Optimization Techniques

- **Heuristic methods** → Approximate solutions
 - **Linear Programming** → Mathematical approach
 - **Greedy algorithms** → Local optimum selection
 - **Dynamic Programming** → Optimal substructure
-

◆ Steps in Optimization

1. Define the problem
 2. Identify objective function
 3. Identify constraints
 4. Apply optimization technique
 5. Evaluate results
-

◆ Advantages

- Improves system efficiency
 - Reduces cost and time
 - Enhances performance
 - Better resource utilization
-

◆ Disadvantages

- Complex process
 - May require high computation
 - Optimal solution not always guaranteed
-



Quick Summary (Exam Tip)

👉 Optimization = Best solution under constraints